SIMATS ENGINEERING

ASSESSMENT TOOL 1

COURSE NAME:MLA03

COURSE CODE: Reinforcement learning for Environment and Agent

COURSE OUTCOME COVERED

CO2: Apply Dynamic Programming methods to solve reinforcement learning problems and derive optimal policies for decision-making tasks. (BL2, BL3)

Assessment Tool Weightage: 35%

Dr G. A. SENTHIL

QUESTIONS: 10 **Total Marks: 50**

Annexure A

Industry Problem Task **Duration: 2.5h**

Code of Conduct:

I K.Neeha(192425245)(Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.

I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student:

Reinforcement Learning Industry Problem

1. Why is Reinforcement Learning (RL) suitable?

Reinforcement Learning is suitable because it enables an intelligent agent to learn the best actions through continuous interaction with the environment. Instead of following fixed rules, the agent improves its decisions by receiving rewards for good actions and penalties for poor actions. This makes RL ideal for dynamic and uncertain real-world environments.

2. Identify the Agent and Environment

Agent: Intelligent software/system that makes decisions.

Environment: The real-world system where the agent performs actions and receives feedback.

Example (Logistics):

- Agent → Delivery Vehicle
- Environment → Road network, traffic, weather, customers

3. Define the State Space

The state represents the current situation observed by the agent.

Example States

- Current location
- Traffic condition
- Battery/Fuel level
- Weather condition
- Destination status

4. Define the Action Space

Actions are the possible decisions the agent can perform.

Example Actions

- Move forward
- Turn left
- Turn right
- Stop
- Select alternative route
- Recharge (if battery is low)

5. Reward Function

Action	Reward
Successful task completion	+100
Correct decision	+20
Shortest path	+10
Delay	-10
Collision/Error	-50
Task failure	-100

Positive rewards encourage good behavior, while negative rewards discourage poor decisions.

6. Policy Used

The policy determines how the agent selects actions.

Example policies:

- ε-Greedy Policy

- Greedy Policy
- Softmax Policy

In ε-Greedy:

- Explore new actions with probability **ε**.
- Exploit the best-known action with probability $\mathbf{1 - \varepsilon}$.

This balances exploration and exploitation.

7. Learning Process

The agent continuously interacts with the environment.

1. Observe current state.
2. Select an action using the policy.
3. Perform the action.
4. Receive reward.
5. Observe next state.
6. Update the value function or Q-table.
7. Repeat until the optimal policy is learned.

Exploration

Trying new actions to discover better solutions.

Exploitation

Using the best action learned so far to maximize reward.

8. Continuous Learning

Continuous learning allows the agent to:

- Adapt to changing environments.
- Improve decision-making.
- Reduce errors over time.
- Increase efficiency and accuracy.
- Learn from new experiences without manual programming.

9. RL Architecture (Block Diagram)

```
+------------------+
|      Agent       |
+------------------+
        |
    Select Action
        |
        V
+------------------+
|   Environment    |
+------------------+
        |
Next State + Reward
        |
        V
+------------------+
| Update Knowledge |
| (Q-Table/Policy) |
+------------------+
        |
      Repeat
```

10. Real-World Industry Example

Amazon Logistics

Amazon uses AI and Reinforcement Learning to optimize delivery routes, reduce fuel consumption, improve delivery speed, and efficiently allocate delivery vehicles based on traffic and customer demand.

(For other applications, replace Amazon with the relevant company, e.g., Netflix for recommendations, Uber for transportation, Google DeepMind for energy management.)

RUBRICS FOR CALCULATION

Criteria	**Weightage**	**Level 4 – Exemplary**	**Level 3 – Proficient**	**Level 2 – Developing**	**Level 1 – Beginning**
Criteria	Weightage	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning
Understanding of Industry Problem	20%	Comprehensive understanding of real-world problem and constraints	Good understanding with minor gaps in requirements	Basic understanding; some requirements unclear	Poor understanding; misinterprets problem context
Application of Engineering Concepts	25%	Applies advanced and relevant concepts effectively to solve the problem	Applies appropriate concepts with minor errors	Limited application of concepts	Incorrect or no application of concepts
Solution Design	25%	Innovative, practical, and feasible solution aligned with industry standards	Logical and feasible solution with minor gaps	Basic solution with limited feasibility	Unclear or impractical solution
Use of Tools	15%	Effective use of modern tools, technologies, and real data	Appropriate use of tools with correct implementation	Limited or inconsistent use of tools	Minimal or incorrect use of tools
Analysis	10%	Thorough analysis with validated results and performance evaluation	Good analysis with reasonable validation	Basic analysis with limited validation	Poor or no analysis
Professionalism	5%	Highly professional, well-documented, and clearly presented work	Good documentation and presentation	Basic documentation with some gaps	Poor documentation and presentation